

YISHUN JUNIOR COLLEGE
Mathematics Department

SOLUTION

Examination : Preliminary Examination

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Subject : JC2 H2 Mathematics

Paper No. : 2

Q ⁿ	Key Steps / Solution
1	$\ln(\cos x)$ $= \ln\left(1 - \frac{(x)^2}{2!} + \frac{(x)^4}{4!} - \frac{(x)^6}{6!} + \dots\right)$ $= \ln\left(1 + \left(-\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots\right)\right)$ $= \left(-\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots\right) - \frac{1}{2}\left(-\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots\right)^2 + \frac{1}{3}\left(-\frac{x^2}{2} + \dots\right)^3 \dots$ $= \left(-\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots\right) - \frac{1}{2}\left(\frac{1}{4}x^4 - \frac{1}{24}x^6 + \dots\right) + \frac{1}{3}\left(-\frac{1}{8}x^6 + \dots\right) \dots$ $= -\frac{1}{2}x^2 - \frac{1}{12}x^4 - \frac{1}{45}x^6 + \dots$ Putting $x = \frac{\pi}{3}$, $\ln\left(\cos\left(\frac{\pi}{3}\right)\right) = -\frac{1}{2}\left(\frac{\pi}{3}\right)^2 - \frac{1}{12}\left(\frac{\pi}{3}\right)^4 - \frac{1}{45}\left(\frac{\pi}{3}\right)^6 + \dots$ $\ln\left(\frac{1}{2}\right) = \ln(2^{-1}) = -\ln 2 \approx -\frac{\pi^2}{18} - \frac{\pi^4}{972} - \frac{\pi^6}{32805}$ $\ln 2 \approx \frac{\pi^2}{18} + \frac{\pi^4}{972} + \frac{\pi^6}{32805}.$
2(i)	Let x, y and z be the number of batches of Releve, Xtraease and Influze to be made respectively. $x + 2y + z = 60 \quad \dots (1)$ $x + y + z = k \quad \dots (2)$ $3x + 6y + 2z = 140 \quad \dots (3)$ $3 \times (1) - (3) : z = 40$ (independent of k)

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2(ii)	When $z = 40$, (1) becomes : $x + 2y = 20 \dots (4)$ (2) becomes : $x + y = k - 40 \dots (5)$ (4) – (5) : $y = 60 - k$ Consider $y = 60 - k \geq 0 \Rightarrow k \leq 60$. $2 \times (5) - (4) : x = 2k - 100$ Consider $x = 2k - 100 \geq 0 \Rightarrow k \geq 50$. Hence, $50 \leq k \leq 60$. 2(iii) RREF $\begin{bmatrix} 1 & 2 & 1 & 60 \\ 1 & 1 & 1 & 55 \\ 3 & 6 & 2 & 140 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 40 \end{bmatrix}$ $x = 10, y = 5$ and $z = 40$ Profit, $P = 14x + 10y + 10z$ $= 14(10) + 10(5) + 10(40) = 590$ Hence the profit made = \$590000.
3(a)	$z + 4 - i = 2\sqrt{3}$ Circle center $(-4, 1)$ with radius $2\sqrt{3}$. (b)(i) Sketch P on the Argand diagram in (a).

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(b)(ii)	$\sin \beta = \frac{2\sqrt{3}}{4}$, $\beta = \sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$ Half line P : $\arg(z - i) = -\frac{2\pi}{3}$. Using geometric approach, $\sin\left(\frac{\pi}{3}\right) = \frac{a}{2\sqrt{3}}$, $\cos\left(\frac{\pi}{3}\right) = \frac{b}{2\sqrt{3}}$ $a = 3$, $b = \sqrt{3}$ $\therefore x = -1$, $y = 1 - \sqrt{3}$ The point of contact of P and C is $z = -1 + i(1 - \sqrt{3})$. Exact value of $m = \text{gradient of line } P = \frac{\sqrt{3}}{1} = \sqrt{3}$
4 (a)(i)	Consider $\begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 9 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ 16 \\ 48 \end{pmatrix} = 16 \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$ A vector which is parallel to both planes p_1 and p_2 is $\begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$. (a)(ii) Let $y = 0$, $5x + z = 2$ $x - 3z = -6$ Solving, $x = 0$, $z = 2$. Therefore, $(0, 0, 2)$ is common to both planes. l_1: $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$ where λ is a real parameter. l_1: $y = \frac{z-2}{3}, x = 0$

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(b)(i)	If 3 planes, p_1, p_2 and p_3 intersect in a $l_1 \Rightarrow l_1$ lies on p_3. p_3 must contain $\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$: $\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ a \\ b \end{pmatrix} = 6 \Rightarrow b = 3$ Normal vector of p_3 is perpendicular to l_1: $\begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ a \\ b \end{pmatrix} = 0 \Rightarrow a + 3b = 0$ If $b = 3$, then $a = -9$.
(b)(ii)	Consider $\frac{2}{\sqrt{5^2 + 3^2 + 1^2}} = \frac{6}{\sqrt{1^2 + a^2 + b^2}}$ $\Rightarrow \frac{6}{3\sqrt{35}} = \frac{6}{\sqrt{1 + a^2 + b^2}}$ $\Rightarrow 3\sqrt{35} = \sqrt{1 + a^2 + b^2} \Rightarrow \sqrt{315} = \sqrt{1 + a^2 + b^2} \Rightarrow a^2 + b^2 = 314$
5 (a) (i)	$v_n - v_{n+1}$ $= \frac{(n+3)}{n(n+1)(n+2)(n+3)} - \frac{(n+4)}{(n+1)(n+2)(n+3)(n+4)}$ $= \frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)}$ $= \frac{(n+3) - n}{n(n+1)(n+2)(n+3)} = \frac{3}{n(n+1)(n+2)(n+3)} = 3u_n, \text{ So } k = 3.$
(a) (ii)	$\sum_{n=1}^N u_n = \frac{1}{3} \sum_{n=1}^N (v_n - v_{n+1}) = \frac{1}{3} (v_1 - v_2$ $+ v_2 - v_3$ $+ v_3 - v_4$ $+ \dots$ $+ v_N - v_{N+1})$ $= \frac{1}{3} (v_1 - v_{N+1})$ $\sum_{n=1}^N u_n = \frac{1}{3} ((1+3)u_1 - (N+4)u_{N+1})$ $= \frac{1}{3} \left(\frac{4}{(1)(2)(3)(4)} - \frac{(N+4)}{(N+1)(N+2)(N+3)(N+4)} \right)$ $= \frac{1}{3} \left(\frac{1}{(1)(2)(3)} - \frac{1}{(N+1)(N+2)(N+3)} \right)$ $= \frac{1}{3} \left(\frac{1}{3!} - \frac{N!}{(N+3)!} \right)$

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(b)	Let P_n be the statement $\sum_{r=3}^n r3^{r+2} = \frac{1}{4} \left((2n-1)3^{n+3} - 729 \right)$. When $n = 3$: $\text{LHS} = \sum_{r=3}^3 r3^{r+2} = 3(3^5) = 3^6 = 729$ $\text{RHS} = \frac{1}{4} \left(5(3^6) - 729 \right) = 729 = \text{LHS}$ Hence, P_3 is true. Assume P_k is true for some $k \in \mathbb{Z}^+$, $k \geq 3$. That is, $\sum_{r=3}^k r3^{r+2} = \frac{1}{4} \left((2k-1)3^{k+3} - 729 \right)$. Want to show P_{k+1} is also true. That is, $\sum_{r=3}^{k+1} r3^{r+2} = \frac{1}{4} \left((2k+1)3^{k+4} - 729 \right)$. $\begin{aligned} \text{LHS} &= \sum_{r=3}^k r3^{r+2} + (k+1)3^{k+3} \\ &= \frac{1}{4} \left((2k-1)3^{k+3} - 729 \right) + (k+1)3^{k+3} \quad (\text{using assumption}) \\ &= \frac{1}{4} \left((2k-1)3^{k+3} - 729 + 4(k+1)3^{k+3} \right) \\ &= \frac{1}{4} \left(3^{k+3} ((2k-1) + 4(k+1)) - 729 \right) \\ &= \frac{1}{4} \left(3^{k+3} (6k+3) - 729 \right) = \frac{1}{4} \left((2k+1)3^{k+4} - 729 \right) = \text{RHS} \end{aligned}$ Hence, P_k is true $\Rightarrow P_{k+1}$ is also true. Since P_3 is true and P_k is true $\Rightarrow P_{k+1}$ is also true, the statement P_n is true for $n \in \mathbb{Z}^+$, $n \geq 3$. $\begin{aligned} \sum_{r=1}^n r3^r &= \frac{1}{9} \sum_{r=1}^n r3^{r+2} \\ &= \frac{1}{9} \left(3^3 + (2)(3^4) + \sum_{r=3}^n r3^{r+2} \right) \\ &= \frac{1}{9} \left(189 + \frac{1}{4} \left((2n-1)3^{n+3} - 729 \right) \right) = \frac{3}{4} + \frac{3}{4} \left((2n-1)3^n \right) = \frac{3}{4} \left(1 + (2n-1)3^n \right) \end{aligned}$

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6(i)	Let X be the r.v. “the number of requests for the booking of a room in the student resource centre per day”. $X \sim \text{Po}(4)$ $P(X > 4) = 1 - 0.6288363 = 0.37116307 = 0.371 \text{ (to 3 s.f.)}$																				
6(ii)	$P(X \leq 4 X > 2) = \frac{P(X \leq 4 \text{ \& } X > 2)}{P(X > 2)}$ $= \frac{P(X = 3) + P(X = 4)}{1 - P(X \leq 2)}$ $= \frac{0.19537 + 0.19537}{1 - 0.2381} = 0.513 \text{ (to 3 s.f.)}$																				
6(iii)	Let Y be the r.v. “the number of days (out of 5) for which not all requests for the booking of a room for that day can be met”. $Y \sim \text{Bin}(5, 0.37116307)$ $P(Y \leq 3) = 0.93328441 = 0.933 \text{ (to 3 s.f.)}$																				
6(iv)	Let the least number of rooms be N. $P(X > N) < 0.05$ $1 - P(X \leq N) < 0.05$ $P(X \leq N) > 0.95$ Using Graphing Calculator, we have <table border="1"> <thead> <tr> <th>N</th><th>$P(X \leq N)$</th></tr> </thead> <tbody> <tr><td>0</td><td>0.0183</td></tr> <tr><td>1</td><td>0.0915</td></tr> <tr><td>2</td><td>0.2381</td></tr> <tr><td>3</td><td>0.4334</td></tr> <tr><td>4</td><td>0.6288</td></tr> <tr><td>5</td><td>0.7851</td></tr> <tr><td>6</td><td>0.8893</td></tr> <tr><td>7</td><td>0.9488</td></tr> <tr><td>8</td><td>0.9786</td></tr> </tbody> </table> From the table, $N = 8$.	N	$P(X \leq N)$	0	0.0183	1	0.0915	2	0.2381	3	0.4334	4	0.6288	5	0.7851	6	0.8893	7	0.9488	8	0.9786
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7(a)	Let X be the r.v. “the number of ClearVision television sets sold (out of 30)”. $X \sim \text{Bin}(30, 0.4)$ Let Y be the r.v. “the number of FreePixel television sets sold (out of 35)”. $Y \sim \text{Bin}(35, 0.5)$ Total profit, $T = 300X + 200Y$ $E(X) = 12, \text{Var}(X) = 7.2, E(Y) = 17.5, \text{Var}(Y) = 8.75.$ Therefore, $E(T) = 300(12) + 200(17.5) = 7100$ $\text{Var}(T) = 300^2(7.2) + 200^2(8.75) = 998000$																				

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7(b)(i)	$X \sim \text{Bin}(30, 0.4)$ Since n is large, $np = 12 > 5$ and $nq = 18 > 5$, $X \sim N(12, 7.2)$ approximately. $Y \sim \text{Bin}(35, 0.5)$ Since n is large, $np = 17.5 > 5$ and $nq = 17.5 > 5$, $Y \sim N(17.5, 8.75)$ approximately. Therefore, $T \sim N(7100, 998000)$ approximately. $P(7000 < T < 8000) = 0.3560472 = 0.356$ (to 3 s.f.)									
(b)(ii)	$300X - 200Y \sim N(100, 998000)$ approximately. $P(300X - 200Y > 200) = P(300X - 200Y > 200) + P(300X - 200Y < -200)$ $= 0.4601324 + 0.38197399$ $= 0.84210639 = 0.842$ (to 3 s.f.) NOTE : When continuity correction is used for (b)(i) and (b)(ii), answers are the same when corrected to 3 s.f.									
8(a)(i)	Number of possible seating arrangements with no restrictions = $(9-1)! = 40320$.									
8(a)(ii)	Number of ways = $(3-1)! \times 2! \times (2!)^4 \times 9 = 576$									
8(b)(i)	Number of ways = ${}^9C_5 \left(\frac{5!}{3! \times 2!} \right) \left(\frac{8!}{2!} \right) = 25401600$									
8(b)(ii)	Number of ways = $\frac{11!}{2! \times 2!} = 9979200$.									
9	<table border="1"><thead><tr><th></th><th>JC1 Students</th><th>JC2 Students</th></tr></thead><tbody><tr><td>Male</td><td>$(420/1600) \times 50 = 13.125 = 13$</td><td>$(380/1600) \times 50 = 11.875 = 12$</td></tr><tr><td>Female</td><td>$(450/1600) \times 50 = 14.0625 = 14$</td><td>$(350/1600) \times 50 = 10.9375 = 11$</td></tr></tbody></table> Based on the proportion given, we will select 13 JC1 male students from 420 JC1 male students, 14 JC1 female students from 450 JC1 female students, 12 JC2 male students from 380 JC2 male students, and 11 JC2 female students from 350 JC2 female students. All selections are done using simple random sampling.		JC1 Students	JC2 Students	Male	$(420/1600) \times 50 = 13.125 = 13$	$(380/1600) \times 50 = 11.875 = 12$	Female	$(450/1600) \times 50 = 14.0625 = 14$	$(350/1600) \times 50 = 10.9375 = 11$
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9(i)	Unbiased estimate of the population mean = $\frac{438}{50} = 8.76$ Unbiased estimate of the population variance, $s^2 = \frac{1}{49} \left(\sum (x-9)^2 - \frac{(\sum (x-9))^2}{50} \right)$ $= \frac{1}{49} \left(45 - \frac{(\sum x - 50 \times 9)^2}{50} \right)$ $= \frac{1}{49} \left(45 - \frac{(438 - 450)^2}{50} \right)$ $= 0.8595918367 = 0.860$ (to 3 s.f.)									

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9(ii)	$H_0 : \mu = 9$ $H_1 : \mu < 9$ Under H_0, the test statistic is $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0,1) \text{ approximately by CLT,}$ where $\mu = 9$, $s = \sqrt{0.8595918367}$, $\bar{x} = 8.76$ and $n = 50$. Level of significance : $5\% = 0.05$. From Graphing Calculator, we have $z = -1.8304173$ $p\text{-value} = 0.03359378$ Since the $p\text{-value } 0.03359378 < 0.05$, we reject H_0. Hence, at 5% level of significance, there is sufficient evidence to conclude that the student council has overstated the average number of hours spent on studying per day by a student. Probability of concluding that the mean studying time is less than 9 hours when it is actually 9 hours is 0.05. $H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$ Under H_0, the test statistic is $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0,1) \text{ approximately by CLT,}$ where $\mu = \mu_0$, $s = \sqrt{0.8595918367}$, $\bar{x} = 8.76$ and $n = 50$. Level of significance : $5\% = 0.05$. Critical region : $z > 1.96$. To reject H_0, test-value $\frac{\bar{x} - \mu}{s/\sqrt{n}} < -1.96$ or $\frac{\bar{x} - \mu}{s/\sqrt{n}} > 1.96$. $\Rightarrow \mu_0 < 8.50 \quad \text{or} \quad \mu_0 > 9.02$
10(i)	$P(A \cup B) = 0.5$, and $P(A) + P(B) = 0.7 \neq 0.5$. Hence, A and B are not mutually exclusive.
10(ii)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $0.5 = 0.4 + 0.3 - P(A \cap B)$ $\Rightarrow P(A \cap B) = 0.2$ But $P(A) \times P(B) = 0.4 \times 0.3 = 0.12 \neq 0.2$. Hence, A and B are not independent.

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10(iii)	
10(iv)	Probability of A winning = $\frac{1}{2}p + \frac{1}{2}(1-p)\frac{1}{3} + \frac{1}{2}\frac{1}{3}p = \frac{1}{2}p + \frac{1}{6}$. Set probability of A winning = $\frac{1}{2}$. $\Rightarrow \frac{1}{2}p + \frac{1}{6} = \frac{1}{2} \Rightarrow p = \frac{2}{3}.$ Alternatively, we may set probability of A losing = $\frac{1}{2}(1-p)\frac{2}{3} + \frac{1}{2}\frac{1}{3}(1-p) + \frac{1}{2}\frac{2}{3} = \frac{5}{6} - \frac{1}{2}p = \frac{1}{2}$. $\Rightarrow \frac{5}{6} - \frac{1}{2}p = \frac{1}{2} \Rightarrow p = \frac{2}{3}$
11(a)	The least squares regression line of y on x is the line with the minimum sum of the squared vertical distances from each point to the line while the least squares regression line x on y is the line with the minimum sum of the squared horizontal distances from each point to the line. $(\bar{x}, \bar{y})$ is the point through which the estimated regression line of y on x and that of x on y both pass. $r^2 = \left(\frac{\Sigma(x - \bar{x})\Sigma(y - \bar{y})}{\sqrt{(\Sigma(x - \bar{x})^2)(\Sigma(y - \bar{y})^2)}} \right)^2 = \left(\frac{\Sigma(x - \bar{x})\Sigma(y - \bar{y})}{\Sigma(x - \bar{x})^2} \right) \left(\frac{\Sigma(x - \bar{x})\Sigma(y - \bar{y})}{\Sigma(y - \bar{y})^2} \right) = bd.$

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11 (b)(i)	From the regression line y on x, we have $\bar{y} = \frac{68}{7}\bar{x} + \frac{786}{7}$. Since $\bar{x} = 8$, $\bar{y} = \frac{68}{7}(8) + \frac{786}{7} = 190$. $\Rightarrow \frac{160 + \alpha + 229 + \beta + 225 + 136}{6} = 190 \Rightarrow \alpha + \beta = 390 \dots (1)$ $b = \frac{\Sigma(x - \bar{x})\Sigma(y - \bar{y})}{\Sigma(x - \bar{x})^2} = \frac{68}{7} \Rightarrow \left(\Sigma xy - \frac{\Sigma x \Sigma y}{6} \right) = \frac{68}{7} \left(\Sigma x^2 - \frac{(\Sigma x)^2}{6} \right)$ Hence, $\left((6660 + 9\alpha + 7\beta) - \frac{(48)(750 + \alpha + \beta)}{6} \right) = \frac{68}{7} \left(454 - \frac{2304}{6} \right)$ $\Rightarrow ((6660 + 9\alpha + 7\beta) - 8(750 + \alpha + \beta)) = 680 \Rightarrow \alpha - \beta = 20 \dots (2)$ (1) + (2) : $2\alpha = 410 \Rightarrow \alpha = 205$ (1) - (2) : $2\beta = 375 \Rightarrow \beta = 185$
(b)(ii)	When $y = 210$, $7(210) = 68x + 786$ $\Rightarrow x = 10.05882353 = 10.1$ (to 3 s.f.) The predicted value of x is likely to be reliable as estimation is done by interpolation and $r = 0.985$ is close to 1.
(b)(iii)	